[Dark.Tyro] From Protected to Editable - Adaptive Attack on Perturbation-Based Defense #26

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PHOTOS OF MEMBERS

BASELINE REFERENCE

Paper	Code
Cao, B., Li, C., Wang, T., Jia, J., Li, B., & Chen, J. (2023). IMPRESS: Evaluating the Resilience of Imperceptible Perturbations Against Unauthorized Data Usage in Diffusion-Based Generative AI (arXiv:2310.19248). arXiv. https://doi.org/10.48550/arXiv.2310.19248	https://github.com/AAAAAAsuka/Impress

IMPRESS (Imperceptible Perturbation Removal System)

The chosen baseline paper examines the robustness of image-protection methods (Glaze and PhotoGuard) from an attacker's angle. The attack comes in the form of *purification*: it takes a protected image, iteratively optimizes the pixel values of the protected image, and removes the hidden protective perturbation while keeping visual fidelity.

A *purified* image has two goals: keeping similarity and consistency (via the latent diffusion model's encoder-decoder). This method uses LPIPS-based similarity loss and reconstruction-consistency loss, allowing it to create slight changes in removing the abnormalities caused by perturbations.

WHY THIS PROBLEM IS INTERESTING / CHALLENGING

Given an image protected by a perturbation defense system (such as PhotoGuard), can an informed attacker purify the image, weakening its defense and restoring the ability to perform text-guided image editing while maintaining visual quality? This problem is interesting as IMPRESS exposes a vulnerability in this type of defense through the usage of an adaptive perturbation purification attack. The abstract says the method “can be used as an evaluation platform for future methods”, but the available source code does not deliver this, as it provides two separate hardcoded scripts for two separate experimental setups (glaze_pur.py, pg_mask_pur_helen.py). This creates a practicality and reproducibility gap, which gives us room for improvement. Ideally, IMPRESS should be able to take any protected image (using any method) and run IMPRESS to purify it through a general interface to satisfy this claim.

At the same time, this problem is challenging to address because image-perturbation purification involves several competing objectives that must be satisfied simultaneously:

- **Visual fidelity:** the purified image should remain visually close to the original image before the defensive perturbation was applied.
- **Restored Text-to-Image utility:** the purification process must remove sufficient defensive signals to recover the image’s usability for Text-to-Image generation or editing.
- **Practical feasibility:** the method should operate under realistic constraints, including GPU availability, runtime, number of iterations, and computational budget.
- **Generalizability:** ideally, the approach should remain effective across different perturbation-based defense mechanisms.

These objectives introduce inherent trade-offs. Improving one objective can directly compromise another. For example, applying stronger denoising may more effectively weaken the defensive perturbation and improve Text-to-Image utility, but may simultaneously remove fine-grained visual details, reducing fidelity to the original image.

Therefore, the effectiveness of purification cannot be justified by a strong result in any single dimension. The project evaluates purification as a multi-objective trade-off between **visual fidelity, restored Text-to-Image utility, computational cost, and robustness across defense settings**. A successful method must demonstrate a reasonable balance across these dimensions rather than optimizing one at the expense of the others.

COLAB LINK

https://colab.research.google.com/drive/1X4nUlI3VlmJSVkrwgPuovGthvAhiVP_k?usp=sharing

OTHER REFERENCES (USED IN COLAB)

	Paper	Code

Stable Diffusion System	**Stable Diffusion v1.5 inpainting**: text-guided image editing, the same editing pathway as this theme's InstructPix2Pix sample paper (arXiv:2211.09800)	huggingface.co/stable-diffusion-v1-5
Defense System (Obstacle)	**PhotoGuard**: *Raising the Cost of Malicious AI-Powered Image Editing*, ICML 2023 — arXiv:2302.06588	github.com/MadryLab/photoguard

CHALLENGE: BUILDING A SHIELD WE CAN'T WASH OFF

Our attack acts like a solvent, not a lock-pick. IMPRESS does not inspect the perturbations it removes; it simply optimizes the image until the autoencoder agrees with itself. As a defender, you do not need to disclose what type of protection your image is protected by; whether it be PhotoGuard, Glaze, Mist, or your own custom method. But we do restrict this challenge to only visually imperceptible, image-space perturbation defense models compatible with our fixed text-guided image-editing pipeline.

How to enter:

1. You will be given a pack containing:
    1. 10 PNG clean faces at 512x512 and their inpainting masks
    2. Colab link (Functional Pipeline)
        1. A working PhotoGuard protector you can run in one cell - change one parameter and you have a valid entry (optional use - you are free to use your own image-protection system)
2. Run the image-perturbation-based defense method on the provided 10 PNGs
3. Send back 10 PNGs (512x512) with the same filenames to us.

That is the entire challenge procedure. How you protect these images is your business, not ours. After we receive the images, we will run our fixed purifier over them, edit clean / protected/purified using identical settings, and publish two numbers per entry.

Metrics	**Meaning**	**Ideal Expectation**

R_pipe = (SSIM_pur − SSIM_adv) / (1 − SSIM_adv)	how much of the editing pipeline our “wash” has restored. 1.0 = stripped; 0 = your shield held	**low**
LPIPS(protected, clean)	how much of the photograph you spent to get there	**low**

Good Luck!